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The Editors-in-Chief
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Dear Editors,

We submit the manuscript *"Not All Embeddings Are Gaussian: Revisiting Adaptive HNSW Search"* by Pranav Shenvi, M G Shrivardhan Swaroop, Megha Shah (PES University), Kiran Hombal (University of Illinois Urbana-Champaign) and Shylaja S S (PES University) for consideration as a research article in The VLDB Journal.

The paper is an experimental study of per-query search breadth in graph-based nearest-neighbor search. Ada-ef (SIGMOD 2026) chooses the search breadth (*ef*) of each HNSW query from a score that assumes a query's similarities to the database are Gaussian. We test that assumption query by query on 41 public datasets and re-evaluate the method end to end on 20 of them. The baseline is a hindsight-tuned fixed *ef*, matched on both mean and worst-case recall. The main findings are:

- The Gaussian departs significantly on nine datasets, six of them learned image, audio and recommendation embeddings, and the data type does not predict which.
- The departure hurts Ada-ef's ranking of queries less than its size suggests. The score collapses only on the most extreme datasets, and in controlled data where the hard queries move away from the bulk.
- What limits Ada-ef in practice is its fixed per-query overhead: a 1025-distance probe and a $d \times d$ variance computation. It is slower than the tuned fixed *ef* in 28 of 34 runs. A sweep of probe lengths for both methods on 17 datasets separates the two costs.
- A nonparametric variant that reads its thresholds from the data (PercEF) ranks queries about as well at equal probe length, and is cheaper than the tuned fixed *ef* in 32 of 34 runs.

The study also compares DARTH, LAET and Patience in Proximity. It checks robustness to target recall, k, HNSW degree, random seed and metric. It reports seven pitfalls in evaluating adaptive search, including one that affected our own first evaluation of Ada-ef, and ends with guidelines for practitioners.

We believe the paper fits The VLDB Journal's interest in thorough experimental analyses of data management techniques. Its length (17 pages in the journal's two-column template) allows the full per-dataset results that a conference version would leave out. All code, run scripts and the result files behind every table and figure are available at `https://anonymous.4open.science/r/PercEF/`; every number can be regenerated from them in minutes.

This manuscript is original, has not been published, and is not under consideration for publication elsewhere. All authors have approved the submission. The authors declare no competing interests.

Sincerely,

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authors)
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